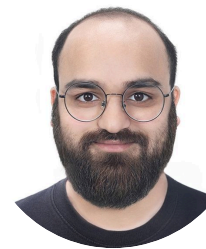


Muhammad Umar Graduate Research Assistant

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PROFILE

I am a combined MS–PhD candidate and BK21 Research Fellow at the University of Ulsan, South Korea, specializing in AI for industrial reliability. My research combines signal processing and deep learning to develop accurate, interpretable, and physics-informed methods for fault diagnosis, predictive maintenance, and Remaining Useful Life prediction using vibration and acoustic emission signals. I have authored more than 25 peer-reviewed journal and conference publications, including papers in IEEE Access, Scientific Reports, and Engineering Failure Analysis. My work has been recognized with three international Best Paper/Presentation Awards and two MDPI Sensors Editor's Choice recognitions. I am passionate about building trustworthy, data-efficient AI solutions that can be reliably deployed in safety-critical industrial applications.

RESEARCH INTERESTS

- Machine Learning and Deep Learning for Real-World Applications
- Signal Processing and Time-Frequency Analysis for Intelligent Systems
- Temporal Representation Learning
- Fault Diagnosis, Prognostics, and Remaining Useful Life (RUL) Prediction
- Explainable AI and Trustworthy Machine Learning
- Vision Transformers and Generative AI for Industrial or Healthcare AI
- Physics-Informed AI and Graph Neural Networks for Smart Sensing
- Multimodal and Low-Sample Learning for Condition Monitoring

EDUCATION

MS-PhD (Combined) AI and Computer Engineering 2024 – Present | South Korea
University of Ulsan

BK21 Research Fellow at the Ulsan Industrial AI (UIAI) Lab focusing on physics-informed deep learning, advanced signal processing, and predictive maintenance for industrial machinery.

BS Computer Systems Engineering 2019 – 2023 | Pakistan
University of Engineering and Technology Peshawar

Focused on system programming, digital signal processing, and machine learning. Completed a Final Year Project on Deepfake Video Detection, using deep neural networks to identify synthetic media manipulations and temporal inconsistencies in video streams.

FSc Pre-Engineering 2017 – 2019 | Pakistan
Ideal College Peshawar

Completed higher secondary education as the College Topper (1st Position) and ranked 3rd overall among all engineering students.

Matric Science 2010 – 2017 | Pakistan
Ummah Children Academy

PROFESSIONAL EXPERIENCE

Research Student Ulsan Industrial AI Lab

University of Ulsan

Working on AI-based solutions for real-world industrial problems, focusing on machine learning, deep learning, and signal processing. Involved in research projects related to fault diagnosis, predictive maintenance, and intelligent systems.

Machine Learning Project Intern

Artificial Intelligence in Healthcare

Worked on the prediction of blood-related diseases using AI, particularly RBC which is comprised of an image segmentation process, for which the UNET architecture was used.

Machine Learning Engineer

Ayass BioScience

Applied AI techniques to healthcare data, focusing on developing machine learning models for accurate disease prediction. Led model development, testing, and optimization to improve clinical decision-making and healthcare outcomes.

Software Developer

ERISP Company

Gained hands-on experience with the Odoo framework, working on module customization, testing, and debugging using Python and XML during my internship.

JOURNAL PAPERS PUBLICATIONS

Burst-informed acoustic emission framework for explainable failure diagnosis in milling machines [✉](#)

Engineering Failure Analysis

Failure-consistent degradation state-space modeling of tool wear evolution and remaining useful life in machining systems [✉](#)

Engineering Failure Analysis

Advanced Fault Diagnosis in Milling Machines Using Acoustic Emission and Transfer Learning [✉](#)

IEEE Access

A distribution-level statistical framework for reliable pipeline leak detection using multi-domain signal analysis

Scientific Reports

Milling machine fault diagnosis using acoustic emission and hybrid deep learning with feature optimization [✉](#)

MDPI Applied Sciences

Wavelet Coherence-Aware Multi-Branch Deep Ensemble for Fault Identification in Centrifugal Pumps

Scientific Reports

Physics guided fused image learning with enhanced squeeze excitation for failure analysis of multistage centrifugal pumps [✉](#)

Scientific Reports

Advanced fault diagnosis in milling cutting tools using vision transformers with semi-supervised learning and uncertainty quantification [✉](#)

Scientific Reports

A hybrid deep learning approach for bearing fault diagnosis using continuous wavelet transform and attention-enhanced spatiotemporal feature extraction [↗](#)

MDPI Sensors

Advanced Bearing-Fault Diagnosis and Classification Using Mel-Scalograms and FOX-Optimized ANN [↗](#)

MDPI Sensors

Enhanced Fault Diagnosis in Milling Machines Using CWT Image Augmentation and Ant Colony Optimized AlexNet [↗](#)

MDPI Sensors

Acoustic Emission-Based Pipeline Leak Detection and Size Identification Using a Customized One-Dimensional DenseNet [↗](#)

MDPI Sensors

A Hybrid Deep Learning Framework for Fault Diagnosis in Milling Machines [↗](#)

MDPI Sensors

An Optimized Few-Shot Learning Framework for Fault Diagnosis in Milling Machines [↗](#)

MDPI Machines

Uncertainty-Calibrated Residual Conformal Monitoring of Wind Turbine SCADA Data for Cross-Asset Anomaly Detection Under Distribution Shift [↗](#)

MDPI Information

A Physics-Guided Statistical Framework for Bearing Fault Diagnosis Under Unit Variability and Rotational Speed Uncertainty [↗](#)

MDPI Information

INTERNATIONAL CONFERENCE PAPERS

An Interpretable Lightweight CNN Framework for Fault Diagnosis in Centrifugal Pumps Using Time-Frequency Scalograms [↗](#)

FICTA 2025 (UK)

A Self-Supervised Event-Based Learning Framework for Acoustic Emission-Based Fault Identification in Milling Machines

MLMI 2026 (Japan)

Stockwell Transform and CNN-Based Pipeline Leak Detection Using Sobel-Filtered Acoustic Emission Signals

ICCIIoT 2024 (Pak)

Enhanced Bearing Fault Diagnosis Using Statistical Time Features and Independent Component Analysis: A Com-parative Study of Neural and Non-Neural Models

ICCIIoT 2024 (Pak)

Advanced Fault Diagnosis in Milling Machines Using CQ-NSGT and Deep Learning [↗](#)

FICTA 2025 (UK)

Local and Global Feature Extraction Using Convolutional Autoencoders and Convolution Neural Networks for Diagnosing Milling Machine Faults [↗](#)

ISDIA 2025 (Dubai)

Attention-Guided Dual Feature Extractor for Bearing Fault Diagnosis from Vibration Data

CSCI 2025 (US)

Selective Kernel Domain Adaptation with LMMD Alignment for Milling Machines Fault Diagnosis 
ICRAI 2026 (Pak)

Bearing Fault Diagnosis: Class-Conditional Deep Domain Adaptation for Generalization Across Machines 
ICRAI 2016 (Pak)

Fault Diagnosis of Centrifugal Pumps Using Adaptive Spectral-Temporal Transformer Learning
MLMI 2026 (Japan)

SKILLS

Programming & Tools

Python, MATLAB, Git, LaTeX, Power BI, Google Colab

Machine Learning & Deep Learning

CNNs, LSTMs, Vision Transformers, Physics-Informed ML, Autoencoders, Conformal Prediction

Explainable AI & MLOps

XAI (Grad-CAM, SHAP), MLflow, Weights & Biases (W&B), Model Deployment

Signal Processing & Analysis

Acoustic Emission Analysis, FFT, STFT, CWT, Time-Frequency Analysis, Feature Extraction

Frameworks & Libraries

PyTorch, TensorFlow, Scikit-Learn, Hugging Face

AWARDS AND ACHIEVEMENTS

BK21+ Research Fellowship

Selected under the Brain Korea program for contributing to high-impact AI research. Awarded a fully funded graduate scholarship for academic excellence in Artificial Intelligence and Computer Engineering.

Best Paper Award – FICTA 2025

Received the Best Paper Award at the 11th International Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA 2025) for outstanding contribution to fault diagnosis using lightweight CNN architectures.

Best Presentation Award - MLMI 2026

Awarded Best Presentation at the International Conference on Machine Learning and Machine Intelligence (MLMI 2026, Tokyo, Japan) for presenting a self-supervised event-based learning framework for acoustic emission fault identification.

Best Paper Award - ICRAI 2026

Received the Best Paper Award at the International Conference on Robotics and Artificial Intelligence (ICRAI 2026, NUST, Pakistan) for research on selective kernel domain adaptation with LMMD alignment for milling machine fault diagnosis.

Editor's Choice Award – MDPI Sensors (2026)

Recognized by the editorial board of Sensors (MDPI) for the high impact and novel hybrid deep learning framework developed for milling machine fault diagnosis.

Editor's Choice Award – MDPI Sensors (2025)

Selected for Editor's Choice recognition in Sensors (MDPI) for research on acoustic emission-based pipeline leak detection using customized 1D DenseNet architectures.

CERTIFICATES

- Deep Learning A-Z 2024: Neural Networks, AI & ChatGPT Prize [↗](#)
- Supervised Machine Learning: Regression and Classification [↗](#)
- Specialization Mathematics for Machine Learning and Data Science [↗](#)
- Python for Data Science, AI & Development [↗](#)
- Signal Processing Onramp [↗](#)
- Programming for Everybody (Getting Started with Python) [↗](#)
- Python for Computer Vision with OpenCV and Deep Learning [↗](#)

PROJECTS

Fault Diagnosis & Maintenance

- Pipeline Leak Detection
- Bearing Fault Diagnosis
- Milling Machine Health Monitoring
- Explainable AI models for Centrifugal Pumps

AI & Healthcare

- Disease Prediction System (Ayass BioScience)
- Clinical Doctor Web App

Academic Projects

- Deepfake Detection
- Real Estate Price Prediction

REFERENCES

Professor Jong-Myon Kim, *Professor*, University of Ulsan
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Dr. Muniba Ashfaq, *Lecturer*, University of Engineering and Technology Peshawar, Pakistan
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Dr. Izaz Raouf, *Postdoctoral Researcher*, Case Western Reserve University
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